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Instruments measuring the quality of life among people living with type-2 diabetes mellitus in India: A systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-043831
Article Type:	Protocol
Date Submitted by the Author:	14-Aug-2020
Complete List of Authors:	Demla, Chetna; Manipal Academy of Higher Education, Department of Commerce Thomas, Anns; Manipal Academy of Higher Education, Department of Commerce Jose, Jomol; Manipal Academy of Higher Education, Department of Commerce Joshy, Angela; Manipal Academy of Higher Education, Department of Commerce Hrishikesh, M; Manipal Academy of Higher Education, Department of Commerce Rajendran, Ambigai; Manipal Academy of Higher Education, Department of Commerce Parsekar, Shradha; Manipal Academy of Higher Education, Public Health Evidence South Asia, Department of Health Information,Prasanna School of Public Health (PSPH), Manipal Academy of Higher Education
Keywords:	General diabetes < DIABETES & ENDOCRINOLOGY, DIABETES & ENDOCRINOLOGY, HEALTH ECONOMICS, PUBLIC HEALTH

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Instruments measuring the quality of life among people living with type-2 diabetes mellitus in India: A systematic review protocol

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Abstract

Purpose: Diabetes is perceived as a worldwide epidemic, with the possibility to cause an overall healthcare emergency. Diabetes affects the quality of life (QoL) of an individual and there exists multiple instruments to measure QoL. The purpose of this systematic review is to summarize the existing QoL instruments and to find out the dimensions of the identified instruments among adult type-2 diabetes population in India. Additionally, we plan to determine the psychometric properties of the identified QoL instruments.

Methods and analysis: Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocol guidelines was followed to report this systematic review protocol. Search will be conducted on PubMed (via Medline), Scopus, CINAHL and Web of Science. We will include studies with adult individuals diagnosed with type-2 diabetes mellitus with and without complications and exclude studies with type1 diabetes or other clinical illness. Five authors in pairs will independently screen the articles and extract studies that met inclusion criteria. COSMIN checklist will be used to measure the methodological quality of the included studies. Narrative synthesis will be performed to analyse the findings of the studies.

Ethics and dissemination: Ethical permission is not applicable as this is a systematic review protocol. We intend to present the systematic review findings in conference and publish in peer-reviewed journal.

PROSPERO registration number: CRD42020180432.

Keywords: Type-2 Diabetes; Instruments; Quality of life; Systematic review.

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Article summary

Strengths and limitations

1. To best of our knowledge this is the first systematic literature review to critique the psychometric properties of diabetes specific and generic quality of life measures in India.
2. Consensus Based Standards for the collection of Health Measurement Instruments study will be followed to define the psychometric properties.
3. Considering resource constraints, the search is limited to English, Indian setting and will not cover grey literature.

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Introduction

"Type-2 diabetes mellitus" is a long-standing health condition caused due to insulin resistance. It was hitherto known as "non-insulin-dependent" or "adult-onset diabetes". [1] Initial signs of type-2 diabetes may include repeated urination, increased appetite and thirst, blurred vision, fatigue, sluggish healing of cuts and wounds, which may go unnoticed. Resulting in diagnosing the disease many years after onset, when disease has progressed to complications. Until recently, type-2 diabetes was mainly diagnosed among adults, but now its incidence is found to be intensifying in youth and children.[1]

Diabetes is perceived as a worldwide epidemic, with the possibility to cause an overall healthcare emergency. It has been anticipated that by 2025, approximately 300 million individuals would be affected by it. [2] About 50.9 million Indians experience ill effects of diabetes and by 2025 India will be the "diabetes capital in the world" with estimated 80 million diagnosed with it.[2] Nevertheless, India Today (2019) has announced that India already has the maximum estimated number of people diagnosed with diabetes in the world. In 2006, there were about 40.9 million Indians with diabetes, out of which 90 percent were type-2 diabetes, with varying prevalence in rural (2.4%) and urban (11.6%) regions.[3] There are multiple factors responsible for growing burden of diabetes viz. unhealthy diets, sedentary lifestyles, rapid urbanization, substance use, improvement in life expectancy[1]and excess body weight.[4] Growing cases of diabetes is putting strain on already weak healthcare sector and economy of India.[5]

It is now well-established that diabetes has an effect on quality of life (QoL) of an individual, especially those who live in developing countries.[6] The World Health Organization (WHO) (2019) defines QoL as; "an individual's perception of their position in life in the context of the culture and

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value systems in which they live and in relation to their goals, expectations, standards and concerns". QoL encompasses person's overall health, opinions, social interactions and association to various characteristics of one's environment thus making a multidimensional concept.[7] The "health-related quality of life" (HRQoL) has been defined within the WHO QoL system as those aspects of QoL that affect person's physical and mental well-being.[8] Considering the importance of QoL, increasingly, across the world, payers are paying closer attention to the QoL of an individual as a result of diseases and trying to optimise QoL within the healthcare budget. To come up with effective interventions it becomes imperative to assess the HRQoL or QoL. Developing a robust understanding of HRQoL or QoL burden of diabetes would help healthcare providers to offer interventions and treatment choices that minimises the impact of diabetes on QoL. However, the selection of QoL and/or HRQoL instruments is a rigorous task and must follow certain standards.

Fitzpatrick and colleagues (1998) have developed the criteria to assess the standard of HRQoL measure and have reported following important issues in measuring HRQoL; whether it is objective or subjective, generic or disease specific, its dimensionality, reported by self or proxy, measurement properties and the HRQoL selection criteria.[9] Assessment of validity and reliability of instruments i.e. psychometric properties helps in understanding the quality of the instruments. Thus, awareness of psychometric or measurement properties of the instrument enables the researchers and clinicians to select the appropriate instrument for their research and clinical practice, respectively. "Reliability is the capacity to replicate a consistent result in time and space or from different observers, with aspects relating to coherence, consistency, equivalence and homogeneity".[10]Validity means an instrument tests exactly what it aims to measure.[11]

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Several studies have been conducted to measure QoL in individuals diagnosed with type-2 diabetes in India, but there has been no review to analyse the results from these studies. Every disease has its own natural history and is influenced by the individual's surroundings; as a result, QoL is affected by the severity of the condition, culture, socio-economic status and other factors. Therefore, specific factors are important for the individual that may not be captured by the instruments developed in different context. Furthermore, some factors might be redundant and irrelevant to some individuals. Therefore, this systematic review is planned to reduce the gap by summarising the existing QoL instruments used among individuals diagnosed with diabetes. This review will be helpful for researchers, decision makers and economists in selecting appropriate instruments to measure QoL among type-2 diabetes mellitus people living in India. Additionally, this systematic review will try to analyse different aspects of QoL by summarizing the dimensions of the instruments. This review will assess the requirement (if any) for additional measurement properties of the instrument or help in recommending robust instruments for measuring QoL of type-2 diabetes individuals in India.

Methodology

We followed "Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) 2015 guidelines"[12]to report this systematic review protocol.

Inclusion and exclusion criteria

Participants: Adult (>/= 18 years) individuals having type-2 diabetes mellitus (diagnosed by medical practitioner and/or confirmed by laboratory reports) in India, with and without complications will be included. In the case of mixed population, we will consider studies reporting subgroups of population of our interest. We will exclude individuals with type-1 diabetes mellitus, gestational diabetes, life-threatening illness and those who are terminally ill. Caregivers acting as

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a proxy for the diabetic individual to measure the QoL or studies assessing caregiver QoL will not be considered as we want to restrict the studies to patient reported QoL.

Study design: Studies related to QoL instrument development and/or measurement of psychometric properties will be included.

Outcomes: Our primary outcome is to find out the disease specific and generic QoL instruments for individuals diagnosed with type-2 diabetes mellitus. Secondly, we will critique measurement characteristics of identified instruments as measured by the studies. These psychometric or measurement properties include validity and reliability as defined by "Consensus Based Standards for the collection of Health Measurement Instruments" (COSMIN) study.[13]We will also report dimensions or domains of the identified QoL instruments.

Search Strategy

We will undertake a comprehensive and sensitive computerized database search on PubMed (via Medline), Scopus, CINAHL and Web of Science. We will use search terms based on Medical Subject Heading and free-text search terms. The following keywords will be used, "quality of life", "diabetes mellitus", "instruments", "psychometric properties" and "India" (details in Table 1). The example search strategy can be found in Table 1. Additionally, we will conduct a reference search and forward citation tracking of the included studies to identify potential records. The search will be restricted to studies published in English, published from 1990 till June 2020. The initial search will be undertaken in PubMed, which will be customized for other databases. The results will be exported to reference manager software and duplicates will be eliminated. The citations will then be exported to an Excel spreadsheet to undertake screening.

Table 1: Search concepts and key terms

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Quality of life	Instrument	Measurement properties	Diabetes
quality of life, QOL, Qol	Instruments, OMIs, Tools, Measures, Questionnaire, Score, Patient-reported outcome measures, PROMs, Outcome assessment, Outcome score, Scale	Psychometric*. Validation. Reproducibility. Discriminant. Analysis. Internal Consistency. Validity. Coefficient. Correlation. Measurement Error. Standard error of measurement, Reliability	Diabetes Mellitus, Diabet*, T2DM, NIDDM, Glucose intolerance, Hyperglycemia, Hypoglycemia, metabolic diseases
Search will be limited to India			

Example Search strategy (PubMed)

(((((Instrument* OR OMIs OR Tool* OR Measure* OR Questionnaire* OR Score OR "Patient-reported Outcome Measures" OR PROMs OR "Outcome assessment" OR "Outcome score" OR Scale)) OR ((Psychometric* OR Validation OR Reproducibility OR Discriminant OR Analysis OR "Internal Consistency" OR Validity OR Coefficient OR Correlation OR "Measurement Error" OR "Standard error of measurement" OR Reliability))) AND (("Diabetes

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Mellitus"[Title/Abstract] OR Diabet*[Title/Abstract] OR T2DM[Title/Abstract] OR NIDDM[Title/Abstract] OR "Glucose intolerance"[Title/Abstract] OR Hyperglycemia[Title/Abstract] OR Hypoglycemia[Title/Abstract] OR "metabolic disease*")[Title/Abstract])) AND (("quality of life"[Title/Abstract] OR QOL[Title/Abstract] OR HrQoL)[Title/Abstract])) AND ((INDIA OR india))

Screening and selection of studies

Five authors (CD, AJ, AT, JJ, HMA), independently, will begin with title screening in groups of two. In case of disagreements, we will discuss till consensus and if consensus is not made, we will contact the third author (SP). We will follow the same process to screen the abstracts and full text. We will use screening protocol as documented in Table 2. We will describe the reasons for exclusion, for all studies excluded by the assessors. The screening process will be documented using the PRISMA flow diagram.

Table 2: Screening protocol

1	**Title and Abstract screening**		
A	Is the study written in English and published after 1990?	If answer to both the components is Yes, then go to B	If not Exclude the study
B	Is it a study using QoL instrument? OR	If answer is Yes for either one or all the components, then go to C.	If not exclude the study.

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	Is it a QoL instrument development study or study measuring psychometric properties? Are you in doubt?		
C	Does the study based on QoL of Type 2 Diabetes Mellitus in India? AND Does the participants under study are above 18? Are you in doubt?	If Yes, then go to D.	If not exclude the study.
D	Does the study mention Generic or Disease specific QoL instrument for Type 2 Diabetes? OR Does the study include psychometric property such as Validity of Reliability for measuring standards of QoL instrument? Are you in doubt?	If Yes, then include the study.	If no exclude the study.
2	**Full Text Screening**		
1A.	Study Design	If the answer to any one is Yes, then go to point no. 2. Go to a	Exclude the study based on study design: if none

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	Is it a survey or case study using QoL instruments for type 2 diabetes patients? OR Is it a study related to QoL instrument development? OR Is it a Study measuring psychometric properties?	point no. 1B for more details	of the options are applicable.
1B	• Check for multiple publications of the same study. We will include all publications but embrace the most extensive one.	If the answer to any one is Yes, then go to point no. 2.	
2.	Population a. Does the population under study have type 2 diabetes mellitus (diagnosed by a medical practitioner and/or confirmed by laboratory reports) in India, with or without complications? b. Does study considered adults (> 18 years) population? c. Does it involve a mixed population, but have reported	If the answer to "a" and "b" is Yes, then go to point no: 3. If you are in doubt in case of point "c" please mark the study for discussion.	Exclude based on population if none of the options are applicable. Exclude adults with type 1 diabetes mellitus, gestational diabetes, life-threatening illness and those who are terminally ill. Caregivers acting as a proxy for the diabetic

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	subgroups of population of our interest?		individual to measure the Qol or studies assessing caregiver Qol will not be considered
3.	Outcomes a. Studies assessing Generic or disease-specific instruments for individuals diagnosed with type-2 diabetes mellitus. b. Studies specifying psychometric characteristics such as reliability and validity (defined by Consensus Based Standards for collection of Health Measurement Instruments, COSMIN) of identified instruments as measured by the studies. c. Studies reporting dimensions or domains of the identified QoL instruments.	If the answer to "a" is Yes, then include the study. Additionally, if answer to "b" or "c" is Yes then include the study.	Exclude study based on outcome if none of the mentioned outcomes are assessed.

Data extraction

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The data collation will be undertaken independently by five authors (CD, AJ, AT, JJ, HMA) in groups of two, to minimize errors and reduce potential biases. We will create a structured data collation form to obtain all relevant information from each included study. The data extraction form will be initially pilot-tested for two to three studies. In the event that the study does not provide all the information needed or missing data for the research, the corresponding authors of those studies will be contacted and in case of no reply after waiting for 15 days, studies will be excluded. Whereas any discrepancy will be solved by having a discussion within and between the groups or contacting the third author (SP) for the systematic review. Table 3 enlists the details that we will extract from each included study.

Table 3: Details of data to be extracted

Article information	First Author and year of publication Title Country of origin Citation details Study period
Instrument details	Name of the instrument Outcome measure: Generic or disease specific Timing of administration Instrument use Language used Number and type of dimensions Number of items

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	Scale design Population information: stage of illness, types of complications (if any), age, gender, ethnicity, place of residence, setting etc. Other contextual factors such as region
Psychometric characteristics	Validity Theoretical/conceptual framework Type of validity tests conducted and results Reliability Type of tests conducted and results Statistical Response Sample size Content validity outcomes Time required to complete the instrument Mode of administration (that is self-completion) Acceptability by clinical teams and managers
Author's conclusion	

Quality appraisal of included studies

We will apply the "COSMIN Risk of Bias checklist" to assess quality appraisal of the included studies.[14-16]We will provide an overall quality rating (strong, moderate, limited, conflicting and unknown) to appraise the level of evidence. COSMIN checklist consists of 4-level ratings ("very

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good, adequate, doubtful and inadequate") for each psychometric property. These properties are measurement error, internal reliability, content, construct and structural validity, and hypothesis testing or responsiveness. Each psychometric property constitutes a separate box and consists of 5-10 questions. A methodological score will be given to the responses within the individual psychometric property/ box. To assess the final result, the lowest score i.e. "worst score counts" of the standard box would be considered. E.g. for the content validity, if one item is coded as "inadequate", the overall quality appraisal of content validity study will be coded as 'inadequate'.[14-16]Each study will be assessed by five authors (CD, AJ, AT, JJ, HMA) independently in groups of two and in case of any discrepancy third author (SP) will be contacted.

Description of studies and analysis

We will undertake narrative synthesis to analyse the findings. Generic and disease specific QoL instruments will be listed independently in a tabular form with respective dimensions and target population. Methodological quality of the study will be considered while generating the evidence. Descriptive analysis of various aspects of the methods utilized for testing psychometric properties of QoL instruments will be given to facilitate their comparison and the abstracted data of instruments will be recorded in a table. The narrative synthesis will enable the selection of the most appropriate instrument for type-2 diabetes QoL score for policy making, economic evaluation and future research in India.

Patient and public involvement: We did not involve patient or public involvement for designing and writing of this systematic review protocol.

Ethics and dissemination: Ethical permission is not applicable as this is a systematic review protocol. We intend to present the systematic review findings in conference and publish in peer-reviewed journal.

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Acknowledgment: We are grateful to Manipal Academy of Higher Education (MAHE), Manipal for providing logistic support and infrastructure to carry out this systematic review. We would like to thank the faculties of PHESA, Dept. of Health Information, PSPH, MAHE, Manipal who have helped us in contributing their valuable suggestions and opinions. We would like to express our gratitude to the faculties of the Department of Commerce, MAHE, Manipal for their guidance and encouragement, which has helped us in our work. Our respect and admiration to Mr. Subrata Bhattacharya, Pharma Quant, Kolkata, West Bengal for his worthwhile inputs.

Author contributions: Conceptualization: Chetna Demla, Ambigai Rajendran, and Shradha S. Parsekar; Writing-original draft preparation: Chetna Demla, Anns Thomas, Jomol Jose, Angela L Joshi, and Hrishikesh M A; Writing- review and editing: Shradha S. Parsekar; and Supervision: Chetna Demla and Shradha S. Parsekar. All authors read and approved the final manuscript.

Funding: This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing interest: None of the authors have conflict of interest to declare.

Data Statement: This is a systematic review protocol and we have not started the review, hence, currently, we do not have any data available. However, it can be shared, later once we begin the review, on request by corresponding author.

Word count and no. of tables: main file- 1900 (excluding title page, abstract, strength & limitations, tables and references), abstract-216, strengths and limitations: 74 (3 points), and 3 tables.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	Yes, Page no. 1, line no. 1, Section: Title
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	PROSPERO - CRD42020180432. Yes, line no. 40, Pg 2.
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Yes, Page no. 1, line no. 3-19
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Yes, Page no. 20, line no. 211-215
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	Yes, Page no. 16, line no. 216-217
Sponsor	5b	Provide name for the review funder and/or sponsor	N/A
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	N/A
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Yes, Page no. 4, line no. 110-116, Section: Introduction
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Yes, Page no. 6, line no. 116-121 for Objectives, Line no. 128-142, Section: Methodology for PICO
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years	Yes, Page no. 6-7, line no. 127-142, Section: Inclusion and exclusion criteria

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		considered, language, publication status) to be used as criteria for eligibility for the review	
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Yes, Page no. 7, line no. 143-150, Section: Search strategy
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Yes, Page no. 7, line no. 150-153, Section: Search strategy, and table 1
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Yes, Page no. 9, line 151-153, Section: Search strategy
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	Yes, Page no. 9, line no. 157-162, Section: Screening and selection of studies and Table 2
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Yes, Page no. 13, line no. 165-174, Section: Data extraction
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	Yes, Table 3
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Yes, Page no. 7, line no. 137-142, Section: Outcomes
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Yes, Page no. 15, line no. 177-189, Section: Quality appraisal of included studies
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	N/A
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	N/A
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	N/A
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	Yes, Page no.16, line no. 190-198, Section: Description of studies and analysis
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	N/A

Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	N/A

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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Instruments measuring the quality of life among people living with type-2 diabetes mellitus in India: A systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-043831.R1
Article Type:	Protocol
Date Submitted by the Author:	23-Dec-2020
Complete List of Authors:	Demla, Chetna; Manipal Academy of Higher Education, Department of Commerce Thomas, Anns; Manipal Academy of Higher Education, Department of Commerce Jose, Jomol; Manipal Academy of Higher Education, Department of Commerce Joshy, Angela; Manipal Academy of Higher Education, Department of Commerce Hrishikesh, M; Manipal Academy of Higher Education, Department of Commerce Rajendran, Ambigai; Manipal Academy of Higher Education, Department of Commerce Parsekar, Shradha; Manipal Academy of Higher Education, Public Health Evidence South Asia, Department of Health Information,Prasanna School of Public Health (PSPH), Manipal Academy of Higher Education
Primary Subject Heading:	Diabetes and endocrinology
Secondary Subject Heading:	Health economics, Public health
Keywords:	General diabetes < DIABETES & ENDOCRINOLOGY, DIABETES & ENDOCRINOLOGY, HEALTH ECONOMICS, PUBLIC HEALTH

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Instruments measuring the quality of life among people living with type-2 diabetes mellitus in India: A systematic review protocol

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Abstract

Introduction: Diabetes is a worldwide epidemic, with the possibility to cause an overall healthcare emergency. Diabetes affects the quality of life (QoL) of an individual and there exists multiple instruments to measure QoL. The purpose of this systematic review is to summarise the existing QoL instruments and to find out the dimensions of the identified instruments among adult type-2 diabetes population in India. Additionally, we plan to determine the psychometric properties of the identified QoL instruments.

Methods and analysis: Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocol guideline was followed to report this systematic review protocol. Search will be conducted on Medline (via PubMed, Web of Science), Scopus, and CINAHL. We will include studies with adult individuals diagnosed with type-2 diabetes mellitus with and without complications and exclude studies with type-1 diabetes or other clinical illness. Studies conducted outside India will be excluded. Five authors in pairs will independently screen the articles and extract the data that met inclusion criteria. Consensus-based standards for the selection of health status measurement instruments checklist will be used to measure the methodological quality of the included studies. Narrative synthesis will be performed to analyse the findings of the studies.

Ethics and dissemination: Ethical permission is not applicable as this is a systematic review protocol. We intend to present the systematic review findings in conference and publish in peer-reviewed journal.

PROSPERO registration number: CRD42020180432.

Keywords: Type-2 Diabetes; Instruments; Quality of life; Systematic review.

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Article summary

1. First systematic review proposed to identify and summarise the disease-specific and generic instruments used in India for assessing the quality of life of individuals with type-2 diabetes.
2. Consensus-based standards for the selection of health status measurement instruments study will be followed to define the psychometric properties.
3. Considering resource constraints, the search is limited to English, Indian setting and will not cover grey literature.

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Introduction

"Type-2 diabetes mellitus" is a long-standing health condition that occurs when body is unable to make required amount of insulin or doesn't use insulin appropriately. It is caused due to beta cells dysfunction and insulin resistance. The risk of type-2 diabetes increases with increasing age and therefore, it is diagnosed mostly in middle-aged and older people.[1] However, in the recent times, its incidence is found to be intensifying in youth and children.[1] Initial signs of type-2 diabetes may include repeated urination, increased appetite and thirst, blurred vision, fatigue, sluggish healing of cuts and wounds, which may go unnoticed. As a result of ignorance of these initial symptoms/signs, diabetes may be diagnosed many years after onset, when disease has progressed to complications.

Diabetes is a worldwide epidemic, with the possibility to cause an overall healthcare emergency. It has been anticipated that by 2025, approximately 300 million individuals would be affected by it.[2] About 50.9 million Indians experience ill effects of diabetes and by 2025 India will be the "diabetes capital in the world" with estimated 80 million diagnosed with it.[2] Nevertheless, India already has the maximum estimated number of people diagnosed with diabetes in the world i.e. approximately 73 million. Every fifth individual with diabetes in the world belong to India.[3] In 2006, there were about 40.9 million Indians with diabetes, out of which 90 percent were type-2 diabetes, with varying prevalence in rural (2.4%) and urban (11.6%) regions.[4] There are multiple factors responsible for growing burden of diabetes viz. unhealthy diets, sedentary lifestyles, rapid urbanization, substance use, improvement in life expectancy[1], excess body weight[5] and genetic predisposition.[2] Growing cases of diabetes is putting strain on already weak healthcare sector and economy of India.[6]

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It is now well-established that diabetes has an effect on quality of life (QoL) of an individual.[7] The World Health Organization (WHO) (2019) defines QoL as; "an individual's perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards and concerns". QoL encompasses person's overall health, opinions, social interactions and association to various characteristics of one's environment thus making a multidimensional concept.[8] The "health-related quality of life" (HRQoL) has been defined within the WHO QoL system as those aspects of QoL that affect person's physical and mental well-being.[9] Considering the importance of QoL, increasingly across the world, stakeholders are paying closer attention to the disease-specific QoL of an individual and trying to optimise QoL within the healthcare budget. To come up with effective intervention it becomes imperative to assess the HRQoL or QoL. Developing a robust understanding of HRQoL or QoL burden of diabetes would help healthcare providers to offer effective interventions and treatment choices that minimises the negative impact of diabetes on QoL. However, the selection of QoL and/or HRQoL instruments is a rigorous task and must follow certain standards.

Fitzpatrick and colleagues (1998) have developed the criteria to assess the standard of HRQoL measure and have reported following important issues in measuring HRQoL; whether it is objective or subjective, generic or disease specific, its dimensionality, reported by self or proxy, measurement properties and the HRQoL selection criteria.[10] Assessment of validity and reliability of instruments i.e. psychometric properties helps in understanding the quality of the instruments. Thus, awareness of psychometric or measurement properties of the instrument enables the researchers and clinicians to select the appropriate instrument for their research and clinical practice, respectively. "Reliability is the capacity to replicate a consistent result in time and space

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or from different observers, with aspects relating to coherence, consistency, equivalence and homogeneity".[11] Validity means an instrument tests exactly what it aims to measure.[12]

Several studies[2, 13-16] have been conducted to measure QoL in individuals diagnosed with type-2 diabetes in India, but there has been no systematic review to analyse the results from these studies. A 34-item sensitive, valid and reliable Quality Of Life Instrument for Diabetes patient (QOLID) was developed among middle- and high-income groups.[13] A study conducted in western and southern India found poor QoL among type-2 diabetes attending care centres.[14-15] A study, using modified QoL instrument for Indian diabetic patients, concluded that diabetic retinopathy had negative impact on QoL and the impact worsened with increasing severity of the disease.[16]

Every disease has its own natural history and is influenced by the individual's surroundings. Additionally, culture influences HRQoL or QoL of an individual. Although an instrument may have been linguistically translated into a local language, it may not have the same meaning in that culture.[17] HRQoL is multidimensional concept and incorporates any or combination of negative impact (such as psychological, physical, social, financial) on normal functions of an individual. Considering the chronicity of diabetes, as the time progresses there is heightened risk for complications such as nephropathy, retinopathy, neuropathy, coronary heart disease, and other complications. As a result, there could be variability in terms of self-care behaviours, disease management, and therapeutic adherence. In countries like India, due to lack of health insurance/ financial protection, majority (71%) of healthcare spending is met by out-of-pocket expenditure.[18] The high financial burden may have a larger effect on QoL than disease or its treatment. As a result, QoL is affected by the severity of the condition, culture, socio-economic status and other factors. Therefore, specific factors such as societal issues, community norms, family dynamics, and culture could be some of the important predictors of QoL for Indian individual that may not

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have been captured by the instruments developed in western context. Furthermore, some factors might be redundant and irrelevant to some individuals.

The development and validation of a new HRQoL instrument is a tedious task and there is less evidence available to choose one instrument over another.[17] An ideal instrument for the assessment of QoL in diabetes should be sensitive for changes, provide information about diabetes-specific associations with QoL, and enable comparison between various groups of patients or general population and economic evaluation.[19] Therefore, this systematic review is planned to reduce the gap by summarising the existing QoL instruments used among individuals diagnosed with diabetes. This review will be helpful for researchers, decision makers and economists in selecting appropriate instruments to measure QoL among type-2 diabetes mellitus people living in India. Additionally, this systematic review will try to analyse different aspects of QoL by summarising the dimensions of the instruments. The review will assess the requirement (if any) for additional measurement properties of the instrument or help in recommending robust instruments for measuring QoL of type-2 diabetes individuals in India.

Methodology

We followed "Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) 2015 guidelines" to report this systematic review protocol.[20]

Inclusion and exclusion criteria

Participants: Adult (>/= 18 years) individuals having type-2 diabetes mellitus (diagnosed by medical practitioner and/or confirmed by laboratory reports) in India, with and without complications will be included. In the case of mixed population, we will consider studies reporting subgroups of population of our interest. We will exclude individuals with type-1 diabetes mellitus,

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gestational diabetes, life-threatening illness and those who are terminally ill. Type-2 diabetes individuals with other chronic communicable diseases such as HIV and tuberculosis will be excluded. Studies conducted outside India will be excluded irrespective whether the population were of Indian origin. Caregivers acting as a proxy for the diabetic individual to measure the QoL or studies assessing caregiver QoL will not be considered as we want to restrict the studies to patient reported QoL.

Study design: Studies related to QoL instrument development and/or measurement of psychometric properties will be included.

Outcomes: Our primary outcome is to find out the disease specific (e.g., QOLID)[13] and generic QoL (e.g., EQ-5D, SF-36) instruments for resident Indian individuals diagnosed with type-2 diabetes mellitus. We do not impose a strict definition of QoL and therefore, will include any definition that study uses. Furthermore, if a study has measured a specific domain(s)/construct(s) (e.g., psychological) of QoL it is eligible to be included. The QoL instrument(s) that are developed in other countries are eligible to be included if the studies have validated it among resident Indian population or the instrument was previously validated by other studies. However, included study should cite the appropriate reference that states it has been validated. Secondly, we will critique measurement characteristics of identified instruments as measured by the studies. These psychometric or measurement properties include validity and reliability as defined by "Consensus Based Standards for the collection of Health Measurement Instruments" (COSMIN) study.[21] We will also report dimensions or domains of the identified QoL instruments.

Search Strategy

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We will undertake a comprehensive and sensitive computerized database search on PubMed (via Medline), Scopus, CINAHL and Web of Science. We will use search terms based on Medical Subject Heading and free-text search terms. The following keywords will be used, "quality of life", "diabetes mellitus", "instruments", "psychometric properties" and "India" (details in Table 1). The example search strategy can be found in Table 1. Additionally, we will conduct a reference search and forward citation tracking of the included studies to identify potential records. The search will be restricted to studies published in English, published from 1990 until July 2020. The initial search will be undertaken in Medline (PubMed), which will be customized for other databases. The results will be exported to reference manager software and duplicates will be eliminated. The citations will then be exported to an Excel spreadsheet to undertake screening.

Table 1: Search concepts and key terms

Quality of life	**Instrument**	**Measurement properties**	**Diabetes**
quality of life, QOL, Qol	Instruments, OMIs, Tools, Measures, Questionnaire, Score, Patient-reported outcome measures, PROMs,	Psychometric*. Validation. Reproducibility. Discriminant. Analysis. Internal Consistency. Validity. Coefficient. Correlation. Measurement Error.	Diabetes Mellitus, Diabet*, T2DM, NIDDM, Glucose intolerance, Hyperglycemia, Hypoglycemia, metabolic diseases

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	Outcome assessment, Outcome score, Scale	Standard error of measurement, Reliability	
Search will be limited to India			
Example Search strategy (PubMed) (((((Instrument* OR OMIs OR Tool* OR Measure* OR Questionnaire* OR Score OR "Patient-reported Outcome Measures" OR PROMs OR "Outcome assessment" OR "Outcome score" OR Scale)) OR ((Psychometric* OR Validation OR Reproducibility OR Discriminant OR Analysis OR "Internal Consistency" OR Validity OR Coefficient OR Correlation OR "Measurement Error" OR "Standard error of measurement" OR Reliability))) AND (("Diabetes Mellitus"[Title/Abstract] OR Diabet*[Title/Abstract] OR T2DM[Title/Abstract] OR NIDDM[Title/Abstract] OR "Glucose intolerance"[Title/Abstract] OR Hyperglycemia[Title/Abstract] OR Hypoglycemia[Title/Abstract] OR "metabolic disease*")[Title/Abstract])) AND (("quality of life"[Title/Abstract] OR QOL[Title/Abstract] OR HrQoL)[Title/Abstract])) AND ((INDIA OR India))			

Screening and selection of studies

Five authors (CD, AJ, AT, JJ, HMA), independently, will begin with title screening in groups of two. In case of disagreements, we will discuss until consensus and if consensus is not made, we will contact the third author (SSP). The third author will act as an arbitrator and final decision maker. We will follow the same process to screen the abstracts and full text. We anticipate that our search will yield a large number of patients reported outcome measures, therefore to screen

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the appropriate studies we will use screening protocol as documented in Table 2. We will describe the reasons for exclusion, for all studies excluded by the assessors. The screening process will be documented using the PRISMA flow diagram.

Table 2: Screening protocol

1	**Title and Abstract screening**		
A	Is the study written in English and published after 1990?	If answer to both the components is Yes, then go to B	If not Exclude the study
B	Is it a study using QoL instrument? OR Is it a QoL instrument development study or study measuring psychometric properties? Are you in doubt?	If answer is Yes for either one or all the components, then go to C.	If not exclude the study.
C	Does the study based on QoL of Type 2 Diabetes Mellitus in India? AND Does the participants under study are above 18? Are you in doubt?	If Yes, then go to D.	If not exclude the study.

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D	Does the study mention Generic or Disease specific QoL instrument for Type 2 Diabetes? OR Does the study include psychometric property such as Validity of Reliability for measuring standards of QoL instrument? Are you in doubt?	If Yes, then include the study.	If no exclude the study.
2	**Full Text Screening**		
1A.	Study Design Is it a survey or case study using QoL instruments for type 2 diabetes patients? OR Is it a study related to QoL instrument development? OR Is it a Study measuring psychometric properties?	If the answer to any one is Yes, then go to point no. 2. Go to a point no. 1B for more details	Exclude the study based on study design: if none of the options are applicable.
1B	• Check for multiple publications of the same study. We will include all publications but embrace the most extensive one.	If the answer to any one is Yes, then go to point no. 2.	

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2.	Population a. Does the population under study have type 2 diabetes mellitus (diagnosed by a medical practitioner and/or confirmed by laboratory reports) in India, with or without complications? b. Does study considered adults (> 18 years) population? c. Does it involve a mixed population, but have reported subgroups of population of our interest?	If the answer to "a" and "b" is Yes, then go to point no: 3. If you are in doubt in case of point "c" please mark the study for discussion.	Exclude based on population if none of the options are applicable. Exclude adults with type 1 diabetes mellitus, gestational diabetes, life-threatening illness and those who are terminally ill, diagnosed with HIV or tuberculosis. Caregivers acting as a proxy for the diabetic individual to measure the Qol or studies assessing caregiver Qol will not be considered. Exclude if the study has been conducted outside India, irrespective whether the population were of Indian origin.

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3.	Outcomes a. Studies assessing generic or disease-specific instruments for individuals diagnosed with type-2 diabetes mellitus. b. Studies measuring a specific domain(s)/construct(s) (e.g. psychological) of QoL. c. Studies specifying psychometric characteristics such as reliability and validity (defined by Consensus Based Standards for collection of Health Measurement Instruments, COSMIN) of identified instruments as measured by the studies. d. Studies reporting dimensions or domains of the identified QoL instruments. Note: The QoL instrument(s) that are developed in other countries are eligible to be included if the studies	If the answer to "a" is Yes, then include the study. Additionally, if answer to "b" or "c" is Yes then include the study.	Exclude study based on outcome if none of the mentioned outcomes are assessed.

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	have validated it among Indian population or the instruments are previously been validated by other studies. However, included study should cite the appropriate reference that states it has been validated.		

Data extraction

The data collation will be undertaken independently by five authors (CD, AJ, AT, JJ, HMA) in groups of two, to minimize errors and reduce potential biases. We will create a structured data collation form to obtain all relevant information from each included study. The data extraction form will be initially pilot-tested for two to three studies. In the event that the study does not provide all the information needed or missing data for the research, the corresponding authors of those studies will be contacted and in case of no reply after waiting for 15 days, studies will be excluded. Whereas any discrepancy will be solved by having a discussion within and between the groups or contacting the third author (SSP) for the systematic review. Table 3 enlists the details that we will extract from each included study.

Table 3: Details of data to be extracted

Article information	First Author and year of publication Title Country of origin

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	Citation details Study period
Instrument details	Name of the instrument Outcome measure: Generic or disease specific Timing of administration Instrument use Language used Number and type of dimensions Number of items Scale design Population information: stage of illness, types of complications (if any), age, gender, ethnicity, place of residence, setting etc. Other contextual factors such as region
Psychometric characteristics	Validity Theoretical/conceptual framework Type of validity tests conducted and results Reliability Type of tests conducted and results Statistical Response Sample size Content validity outcomes Time required to complete the instrument

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	Mode of administration (that is self-completion) Acceptability by clinical teams and managers
Author's conclusion	

Quality appraisal of included studies

We will apply the "COSMIN Risk of Bias checklist" to assess quality appraisal of the included studies.[22-24] We will provide an overall quality rating (strong, moderate, limited, conflicting and unknown) to appraise the level of evidence. COSMIN checklist consists of 4-level ratings ("very good, adequate, doubtful and inadequate") for each psychometric property. These properties are measurement error, internal reliability, content, construct and structural validity, and hypothesis testing or responsiveness. Each psychometric property constitutes a separate box and consists of 5-10 questions. A methodological score will be given to the responses within the individual psychometric property/ box. To assess the final result, the lowest score i.e. "worst score counts" of the standard box would be considered. E.g. for the content validity, if one item is coded as "inadequate", the overall quality appraisal of content validity study will be coded as 'inadequate'.[22-24] Each study will be assessed by five authors (CD, AJ, AT, JJ, HMA) independently in groups of two and in case of any discrepancy third author (SSP) will be contacted.

Description of studies and analysis

We will undertake narrative synthesis to analyse the findings. Generic and disease specific QoL instruments will be listed independently in a tabular form with respective dimensions and target population. Furthermore, studies that used instruments developed outside India will be segregated to identify if it were tested among India. Methodological quality of the study will be considered

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while generating the evidence. Descriptive analysis of various aspects of the methods utilised for testing psychometric properties of QoL instruments will be given to facilitate their comparison and the abstracted data of instruments will be recorded in a table. The narrative synthesis will enable the selection of the most appropriate instrument for type-2 diabetes QoL score for policy making, economic evaluation and future research in India.

Our intent is to identify the QoL instruments used among resident Indian diabetes population and compare and contrast the domains/construct of these instruments therefore, restricted the analysis to narrative synthesis. Additionally, we anticipate heterogeneity in population, context, diabetes complications, types of instruments (generic/ condition specific), construct/domains being measured and study methods. For this type of systematic review, statistical pooling might not be appropriate and possible.

Patient and public involvement: We did not involve patient or public involvement for designing and writing of this systematic review protocol. However, considering the importance of stakeholders in research we will explore the possibility of including the stakeholders at the final stage of review. We will share the findings of the final review with at least two patients, researchers/professionals working in the field of quality of life, and diabetes care management, and social workers. We will get the feedback of stakeholders before submitting the final review to the journal. Considering resource constraints, we will select the stakeholders affiliated to the author's institution.

Ethics and dissemination: Ethical permission is not applicable as this is a systematic review protocol. We intend to present the systematic review findings in conference and publish in peer-reviewed journal.

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Acknowledgment: We are grateful to Manipal Academy of Higher Education (MAHE), Manipal for providing logistic support and infrastructure to carry out this systematic review. We would like to thank the faculties of PHESA, Dept. of Health Information, PSPH, MAHE, Manipal who have helped us in contributing their valuable suggestions and opinions. We would like to express our gratitude to the faculties of the Department of Commerce, MAHE, Manipal for their guidance and encouragement, which has helped us in our work. Our respect and admiration to Mr. Subrata Bhattacharya, Pharma Quant, Kolkata, West Bengal for his worthwhile inputs.

Author contributions: Conceptualization: Chetna Demla, Ambigai Rajendran, and Shradha S. Parsekar; Writing-original draft preparation: Chetna Demla, Anns Thomas, Jomol Jose, Angela L Joshi, and Hrishikesh M A; Writing- review and editing: Shradha S. Parsekar; and Supervision: Chetna Demla and Shradha S. Parsekar. All authors read and approved the final manuscript.

Funding: This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing interest: None of the authors have conflict of interest to declare.

Data Statement: This is a systematic review protocol and we have not started the review, hence, currently, we do not have any data available. However, it can be shared, later once we begin the review, on request by corresponding author.

Word count and no. of tables: main file- 2317 (without title page, abstract, strength & limitations, acknowledgement, and other declarations, tables and references), abstract-236, strengths and limitations: 66 (3 points), and 3 tables.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	Yes, Page no. 1, line no. 1, Section: Title
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	PROSPERO - CRD42020180432. Yes, line no. 40, Pg 2.
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Yes, Page no. 1, line no. 3-19
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Yes, Page no. 19, line no. 256-260
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	Yes, Page no. 16, line no. 261-262
Sponsor	5b	Provide name for the review funder and/or sponsor	N/A
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	N/A
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Yes, Page no. 6-7, line no. 112-139, Section: Introduction
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Yes, Page no. 7, line no. 139-141
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design,	Yes, Page no. 7-8, line no. 150-174, Section: Inclusion and exclusion criteria

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		setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Yes, Page no. 9, line no. 176-184, Section: Search strategy
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Yes, Page no. 9, line no. 176-195, Section: Search strategy, and table 1
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Yes, Page no. 10, line 187-153, Section: Search strategy
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	Yes, Page no. 9, line no. 157-162, Section: Screening and selection of studies and Table 2
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Yes, Page no. 15, line no. 198-207, Section: Data extraction
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	Yes, Table 3
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Yes, Page no. 8, line no. 163-174, Section: Outcomes
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Yes, Page no. 17, line no. 209-221, Section: Quality appraisal of included studies
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	N/A
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	N/A
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	N/A
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	Yes, Page no.17-18, line no. 222-237, Section: Description of studies and analysis
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	N/A

Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	N/A

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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BMJ Open

Instruments measuring the quality of life among people living with type-2 diabetes mellitus in India: A systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-043831.R2
Article Type:	Protocol
Date Submitted by the Author:	10-Mar-2021
Complete List of Authors:	Demla, Chetna; Manipal Academy of Higher Education, Department of Commerce Thomas, Anns; Manipal Academy of Higher Education, Department of Commerce Jose, Jomol; Manipal Academy of Higher Education, Department of Commerce Joshy, Angela; Manipal Academy of Higher Education, Department of Commerce Hrishikesh, M; Manipal Academy of Higher Education, Department of Commerce Rajendran, Ambigai; Manipal Academy of Higher Education, Department of Commerce Parsekar, Shradha; Manipal Academy of Higher Education, Public Health Evidence South Asia, Department of Health Information,Prasanna School of Public Health (PSPH), Manipal Academy of Higher Education
Primary Subject Heading:	Diabetes and endocrinology
Secondary Subject Heading:	Health economics, Public health
Keywords:	General diabetes < DIABETES & ENDOCRINOLOGY, DIABETES & ENDOCRINOLOGY, HEALTH ECONOMICS, PUBLIC HEALTH

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Instruments measuring the quality of life among people living with type-2 diabetes mellitus in India: A systematic review protocol

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Abstract

Introduction: Type-2 diabetes mellitus affects an individual's quality of life (QoL); and there are multiple instruments that can be used to measure QoL. The purpose of this systematic review is to identify the existing instruments that have been used to measure QoL in people living with diabetes, and to enlist the major domains (such as physical, and psychological components) available in the identified instruments. Additionally, we plan to determine the psychometric properties of the identified QoL instruments using COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) methodology.

Methods and analysis: The Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocol (PRISMA-P) guideline was followed to report this systematic review protocol. Searches will be conducted on MEDLINE (via PubMed, Web of Science), SCOPUS, and CINAHL. Predetermined inclusion/ exclusion criteria will be applied to the search results, to include studies with adult individuals diagnosed with type-2 diabetes mellitus, with and without complications, and exclude studies with type-1 diabetes or other clinical illness. Studies conducted outside India will be excluded. Five authors in pairs will independently screen the articles and extract the data that meets the inclusion criteria. The COSMIN criteria will be used to assess the risk of bias of included studies. Narrative synthesis will be performed to analyse the findings of the instruments.

Ethics and dissemination: Ethical permission is not applicable, as this is a systematic review. We intend to disseminate the systematic review findings through a national or international conference and publish the findings in peer-reviewed journal.

PROSPERO registration number: CRD42020180432.

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Keywords: COSMIN, Type-2 Diabetes; India; Instruments; Quality of life; Systematic review.

Article summary

1. Systematic review proposed to identify and summarise the disease-specific and generic instruments used in India for assessing the quality of life of individuals with type-2 diabetes.
2. COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) methods will be followed to assess the psychometric properties.
3. We anticipate heterogeneity in population, context, diabetes related complications, types of instruments, domains/themes being measured and study methods.

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Introduction

"Type-2 diabetes mellitus" is one of the non-communicable diseases that occurs when the pancreas is unable to produce required amount of insulin, or when the body cannot use insulin appropriately. The risk of type-2 diabetes rises with increasing age and therefore, it is diagnosed mostly in the middle and older aged population.[1] However, recently its incidence is found to be intensifying among youth and children.[1] Initial signs of type-2 diabetes may include repeated urination, increased appetite and thirst, blurred vision, fatigue, and sluggish healing of cuts and wounds that may remain unnoticed. As a result of lack of awareness of the initial symptoms/ signs compounded with other demand-side and supply-side factors, type-2 diabetes may be diagnosed many years after its onset viz. the stage when the disease has progressed to complications.

Diabetes is a worldwide epidemic, with the possibility to cause an overall healthcare emergency. It has been anticipated that by 2025, approximately 300 million individuals would be affected by it.[2] About 50.9 million resident Indians experience ill effects of diabetes, and by 2025 India will be the diabetes capital of the world with an estimated 80 million diagnosed with the disease.[2] India already has the maximum estimated number of people diagnosed with diabetes globally (approximately 73 million) that indicates every fifth individual with diabetes in the world belongs to India.[3] In 2006, there were about 40.9 million resident Indians with diabetes, out of which 90 percent were type-2 diabetes, with varying prevalence in rural (2.4%) and urban (11.6%) regions.[4] There are multiple factors responsible for the growing burden of type-2 diabetes *viz.* unhealthy diets, sedentary lifestyles, rapid urbanization, substance use, improvement in life expectancy,[1] excess body weight[5] and genetic predisposition.[2] Growing cases of diabetes are straining the presently weak healthcare sector and economy of India.[6]

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It is now well-established that type-2 diabetes affects the quality of life (QoL) of an individual.[7] The World Health Organization (WHO) (2019) defines QoL as; "an individual's perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards and concerns". QoL encompasses a person's overall health, opinions, social interactions and association to various characteristics of one's environment, thus, making it a multidimensional concept.[8] The "health-related quality of life" (HRQoL) has been defined within the WHO QoL system as those aspects of QoL that affect a person's physical and mental well-being.[9] Considering the importance of QoL, stakeholders across the world are increasingly paying attention to the disease-specific QoL of an individual and trying to optimise QoL within the healthcare budget. In such a situation, it becomes imperative to assess the HRQoL or QoL. Developing a robust understanding of HRQoL or QoL among individuals living with type-2 diabetes would help the healthcare providers to offer effective interventions and treatment choices that minimise the negative impact of diabetes on QoL.

Fitzpatrick and colleagues (1998) have developed the criteria to assess the standard HRQoL measure and have reported the following important issues in measuring HRQoL; whether it is objective or subjective, generic or disease specific, its dimensionality, reported by self or proxy, measurement properties and the HRQoL selection criteria.[10] Assessment of validity and reliability of instruments i.e., psychometric properties, helps in understanding the quality of the instruments. Thus, awareness of psychometric or measurement properties of the instrument enables the researchers and clinicians to select the appropriate instrument for their research and clinical practice, respectively. The psychometric properties of the instrument include reliability and validity. "Reliability is the capacity to replicate a consistent result in time and space, or from

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different observers, with aspects relating to coherence, consistency, equivalence and homogeneity".[11] Validity means an instrument tests exactly what it aims to measure.[12]

Studies,[2, 13-16] that investigate the QoL of individuals with type-2 diabetes and the associated factors, have been conducted previously in India. It is challenging to summarize the results of these studies as they used different QoL instruments. The QoL of an individual is influenced by the severity of the disease condition, culture,[17] and other demand-side and supply-side factors. Furthermore, the high financial burden[18] associated with chronic conditions such as diabetes in resource limited countries (like India) may have a larger effect on QoL. As a result, the instruments validated elsewhere may not be able to adequately capture QoL for individuals living in India. Nevertheless, to the best of our knowledge, no systematic reviews have appropriately assessed the psychometric properties of available QoL measures among people living with type-2 diabetes mellitus in India.

Considering the availability of a number of generic and disease-specific questionnaires, it is a challenge to select the most appropriate patient-reported outcome measures (PROMs).[17] An ideal PROM should be sensitive for changes, provide information about disease-specific associations with construct being studied, enable comparison between various groups of patients or general population and be useful in economic evaluation.[19] Therefore, this systematic review is aimed at identifying the existing instruments that have been used to measure QoL in people living with diabetes in India. As the QoL is a multi-dimensional construct, we will enlist the major domains or themes (may include physical, psychological, and social components) of the identified instruments. Additionally, we will determine the psychometric properties (content and structural validity) of the identified QoL instruments using "COnsensus-based Standards for the selection of health Measurement INstruments" (COSMIN) methodology. This review will be helpful for

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researchers, decision makers and economists in selecting appropriate instruments to measure QoL among people living with type-2 diabetes mellitus in India

Methodology

We followed the "Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) 2015 guidelines" to report this systematic review protocol.[20] We will utilize the 10-step procedure to carry out this systematic review as suggested in COSMIN methodology.[21, 22]

Inclusion and exclusion criteria of studies

Participants: Adult (>/= 18 years) individuals having type-2 diabetes mellitus (diagnosed by medical practitioner and/or confirmed by laboratory reports) in India (with and without complications), will be included, irrespective of the duration of diabetes. In the case of mixed population, we will consider studies reporting subgroups of population of our interest. We will exclude individuals with type-1 diabetes mellitus, gestational diabetes, life-threatening illness and those who are terminally ill. Type-2 diabetic individuals with other chronic communicable diseases such as HIV and tuberculosis will be excluded. Studies conducted in countries other than India, with population of Indian origin/ descent, will be excluded. Caregivers acting as a proxy for the diabetic individuals to measure the QoL or studies assessing caregiver QoL will not be considered as we want to restrict to the patient reported QoL.

Study design: Studies related to QoL instrument development and/or measurement of psychometric properties will be included.

Outcomes: The primary outcome is to identify all PROMs that have been used to assess QoL in studies of individuals living in India with type-2 diabetes mellitus. The QoL instruments could be disease specific (e.g., QOLID)[13] or generic QoL (e.g., EQ-5D, SF-36). We do not impose a strict

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definition of QoL and therefore, we will not restrict the inclusion depending on definition of QoL. Considering the multidimensionality of QoL, a study will be eligible to be included if it has measured/ scored a specific domain(s)/ construct(s) (e.g., psychological component) of QoL as a unidimensional measure. The QoL instruments that was developed in other countries are eligible to be included, if it has been validated among resident Indian population living with type-2 diabetes. However, in this case (if required), we will refer to the original PROM development study to look for content of the instrument. Additionally, we will critique measurement properties of identified instruments as measured by the studies. The psychometric or measurement properties in this systematic review will include content validity and internal structure as defined by the COSMIN study. [23-26] Further, justification and description of psychometric properties can be found below under the 'risk of bias assessment of included studies'. We will also report major domains or themes (such as physical or psychological components) of the identified QoL instruments

Search Strategy

We will undertake a comprehensive and sensitive electronic database search on MEDLINE (via PubMed, Web of Science), SCOPUS and CINAHL. Each database will be searched from its inception to March 2021. We will use search terms based on Medical Subject Heading (MeSH) and free-text search terms, and the sensitive search strategy suggested by Terwee *et al*.[27] will be followed. The major search concepts will be, quality of life (construct), diabetes mellitus (population), and instruments. Furthermore, we will apply the measurement property filter as suggested in the COSMIN study.[27] The search concepts and keywords are listed in Table 1. Additionally, we will conduct a reference search and forward citation tracking of the included studies to identify potential records. A designated information scientist will run the search in above-mentioned databases. The search will be restricted to studies published in English language.

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The initial search will be undertaken in PubMed, which will be customized for other databases. The results will be exported to reference manager software where the duplicates will be eliminated. The citations will then be exported to an Excel spreadsheet for screening.

Table 1: Search concepts and key terms

Construct	'quality of life', 'QOL', 'Qol', 'HrQoL', 'HRQL' 'Health related quality of life', 'Quality of life instrument for diabetes patients', 'QOLID', 'EQ-5D', 'SF-36', 'short form 36', 'World Health Organisation Quality of Life Scale-Brief Version', 'WHOQOL-BREF', 'World Health Organization Quality of Life', 'ADDQoL', 'The Audit of Diabetes Dependent Quality of Life', Questionnaire for Life Instrument for Indian Diabetes Patients', etc.
Instrument	'Instruments', 'Tools', 'Measures', 'Questionnaire', 'Score', 'Patient-reported outcome measures', 'PROMs', 'Outcome assessment', 'Outcome score', 'Scale' etc.
Population	'Diabetes Mellitus', 'Diabet*', 'T2DM', 'NIDDM', 'Glucose intolerance', 'Hyperglycemia', 'Hypoglycemia', 'metabolic diseases', 'type-2 diabetes'
Measurement properties	COSMIN measurement properties filter[27] AND exclusion filter[27]
Country	Country specific filter (India)

Screening and selection of studies

Five authors (CD, AJ, AT, JJ, HMA) in groups of two, will begin with title screening independently. In case of disagreements for inclusion or exclusion of studies, a discussion will be held until consensus is achieved. In case a consensus is not achieved, we will contact the third

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author (SSP). The third author will act as an arbitrator and final decision maker. We will follow the same process to screen the abstracts and full text. We anticipate that our search will yield a large number of PROM, therefore, to screen the appropriate studies we will use a screening protocol as documented in Table 2. We will describe the reasons for exclusion for all studies excluded by the assessors. The screening process will be documented using the PRISMA flow diagram.

Table 2: Screening protocol

1	**Title and Abstract screening**		
A	Is the study written in English language and published after 1990?	If answer to both the components is 'yes', then go to B	If not, exclude the study
B	Is it a study using QoL instrument? OR Is it a QoL instrument development study or study measuring psychometric properties? Are you in doubt?	If answer is 'yes' for either one or all the components, then go to C.	If not, exclude the study.
C	Is the study based on QoL of type 2 diabetes mellitus in India? AND Do the participants under study are above or equal to 18 years? Are you in doubt?	If 'yes', then go to D.	If not, exclude the study.

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D	Does the study mention generic or disease-specific QoL instrument for type 2 Diabetes? OR Does the study include psychometric property such as validity or reliability for measuring standards of QoL instrument? Are you in doubt?	If 'yes', then include the study.	If no, exclude the study.
2	**Full Text Screening**		
1A.	Study design Is it a survey or case study using QoL instruments for type 2 diabetes patients? OR Is it a study related to QoL instrument development? OR Is it a study measuring psychometric properties?	If the answer to all is 'yes', then go to point no. 2. Go to a point no. 1B for more details	Exclude the study based on study design: if none of the options are applicable.
1B	Check for multiple publications of the same study. We will include all publications but include the most extensive one.	If the answer to any one is Yes, then go to point no. 2.	

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2.	Population a. Does the population under study have type 2 diabetes mellitus (diagnosed by a medical practitioner and/or confirmed by laboratory reports) in India, with or without complications? b. Does study consider adults (> or = 18 years) population? c. Does it involve a mixed population, but have reported subgroups of population of our interest?	If the answer to "a" and "b" is Yes, then go to point no: 3. If you are in doubt in case of point "c" please mark the study for discussion.	Exclude based on population if none of the options are applicable. Exclude adults with type 1 diabetes mellitus, gestational diabetes, life-threatening illness and those who are terminally ill, diagnosed with HIV or tuberculosis. Caregivers acting as a proxy for the diabetic individual to measure the Qol or studies assessing caregiver Qol will not be considered. Exclude if the study has been conducted outside India, irrespective whether the population were of Indian origin.

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3.	Outcomes a. Studies assessing generic or disease-specific instruments for individuals diagnosed with type-2 diabetes mellitus. b. Studies measuring a specific domain(s)/construct(s) (e.g. psychological) of QoL. c. Studies specifying psychometric characteristics such as reliability and validity (defined by COSMIN) of identified instruments as measured by the studies. d. Studies reporting dimensions or domains of the identified QoL instruments. Note: The QoL instrument(s) that are developed in other countries are eligible to be included if the studies have validated it among Indian population.	If the answer to "a" is Yes, then include the study. Additionally, if answer to "b" or "c" is Yes then include the study.	Exclude study based on outcome if none of the mentioned outcomes are assessed.

192 Data extraction

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The data collation will be undertaken independently by five authors (CD, AJ, AT, JJ, HMA) in groups of two, to minimize errors and reduce potential biases. Any disagreement will be resolved by having a discussion within and between the groups or contacting the third author (SSP) for a final decision. We will create a structured data collation form to obtain all relevant information from each included study. The data extraction form will be initially pilot-tested by all the authors for two to three studies. Table 3 enlists the details that we will extract from each of the included studies. In case of missing data or incomplete information in the study, the corresponding author will be contacted. If there is no reply from the corresponding author after waiting for 15 days, the study will be excluded.

Table 3: Details of data to be extracted

Article information	First Author and year of publication Title Country of origin Citation details Study period
Instrument details	Name of the instrument Outcome measure: Generic or disease specific Timing of administration Instrument use Language used Number and type of dimensions Number of items

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	Scale design Population information: stage of illness, types of complications (if any), age, gender, ethnicity, place of residence, setting etc. Other contextual factors such as region
Psychometric characteristics	Validity Theoretical/conceptual framework Type of validity tests conducted and results Reliability Type of tests conducted and results Statistical Response Sample size Content validity outcomes Time required to complete the instrument Mode of administration (that is self-completion) Acceptability by clinical teams and managers
Author's conclusion	

Risk of bias assessment of included studies

We will apply the COSMIN Risk of bias (RoB) checklist to assess the methodological quality of the included studies.[24, 28, 29] The COSMIN study does not dictate the use of overall checklist and recommends that assessment of each measurement property can be considered as a distinct study. Therefore, for the purpose of this systematic review, we will restrict the checklist to assess content

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validity and internal structure. This corresponds to five checklists of COSMIN RoB checklist: PROM development (35-items), content validity (31-items), structural validity (4-items), internal consistency (5-items) and cross-cultural validity (4-items).[23-25, 28] The content validity echoes the construct that is supposed to be measured and it can affect other measurement properties of the instrument. Therefore, assessment of content validity is essentially the first step of measurement of the validity of a PROM. Overall judgement for content validity will be decided based on 4-point ratings ("very good, adequate, doubtful and inadequate"). To assess the final result, the lowest score i.e., "worst score counts" of the standard box would be considered. [23-25, 28]

The construct (QoL) that is studied in the current systematic review is a multi-dimensional concept, therefore it is important to know whether the instruments accurately represent the multi-dimensionality of QoL in the individuals living with type-2 diabetes mellitus in India.[23-25,28] Hence, structural validity will be assessed. The RoB checklist for assessing structural validity is composed of four items. The RoB assessment and scoring is similar to content validity, as described above.[23-25, 28]

Additional details of assessing risk of bias for content and structural validity can be found in the COSMIN manual.[23-25] Five authors (CD, AJ, AT, JJ, HMA) independently in groups of two, will assess the RoB for each included study and in case of any discrepancy, a third author (SSP) will be contacted. The quality of the evidence will be graded using the, GRADE approach.[23-29.] Besides, the quality of our systematic review itself will be evaluated using COSMIN checklist.

Description of studies and analysis

We will undertake narrative synthesis to analyse the findings. Generic and disease specific QoL instruments will be listed independently in a tabular form with respective dimensions and target population. Furthermore, studies that used instruments developed outside India will be segregated

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to identify if it were validated among India. Methodological quality of the study will be considered while generating the evidence. We will provide a summary of each criterion for measurement properties of PROM as suggested by Terwee et al.[11] Each property will be rated as positive, negative or intermediate.[11] Through this systematic review, we will provide evidence of the psychometric properties (content validity and internal structure) for each of the PROM included in the study. Furthermore, we will make a recommendation based upon the results of the review as suggested by COSMIN manual.[23-25]

Our intent is to identify the QoL instruments used among resident Indian type-2 diabetes population and compare the domains/ construct of these instruments, therefore, the analysis is restricted to narrative synthesis. Additionally, we anticipate heterogeneity in population, context, diabetes complications, types of instruments (generic/ condition specific), construct/ domains being measured and study methods. Considering the aforementioned reasons, we do not intend to undertake statistical pooling.

Patient and public involvement: We did not have patient or public involvement for designing and writing of this systematic review protocol. However, considering the importance of stakeholders in research we will explore the possibility of including the stakeholders at the final stage of review. We will share the findings of the final review with at least two patients, researchers/ professionals working in the field of quality of life, and diabetes care management, and social workers. We will get the feedback of stakeholders before submitting the final review to the journal. Considering the resource constraints, we will select the stakeholders affiliated to the authors' institution.

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Ethics and dissemination: Ethical permission is not applicable as this is a systematic review protocol. We intend to present the systematic review findings in a national and/or international conference and publish the findings in a peer-reviewed journal.

Acknowledgment: We are grateful to Manipal Academy of Higher Education (MAHE), Manipal for providing logistic support and infrastructure to carry out this systematic review. We would like to thank the faculty of PHESA, Dept. of Health Information, PSPH, MAHE, Manipal who have helped us in contributing their valuable suggestions and opinions. We would like to express our gratitude to the faculties of the Department of Commerce, MAHE, Manipal for their guidance and encouragement, which has helped us in our work. We extend our gratitude and admiration to Mr. Subrata Bhattacharya, Pharma Quant, Kolkata, West Bengal for his worthwhile inputs. We are grateful to Dr. Prachi Pundir, Research Officer, PHESA, PSPH, MAHE, Manipal, for proof-reading the document.

Author contributions: Conceptualization: Chetna Demla, Ambigai Rajendran, and Shradha S. Parsekar; Writing-original draft preparation: Chetna Demla, Anns Thomas, Jomol Jose, Angela L Joshy, and Hrishikesh M A; Writing- review and editing: Shradha S. Parsekar; and Supervision: Chetna Demla and Shradha S. Parsekar. All authors read and approved the final manuscript.

Funding: This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing interest: None of the authors have conflict of interest to declare.

Data Statement: This is a systematic review protocol and we have not started the review, hence, currently, we do not have any data available. However, it can be shared on request to corresponding author once we complete the review.

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Word count and no. of tables: main file- 3213 including tables (without title page, abstract, article summary, acknowledgement, and other declarations, and references), abstract- 253, article summary: 66 (3 points), and 3 tables.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Page no, Line no
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	Yes, Page no. 1, line no. 1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	Yes, line no. 43
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Yes, line no. 3-19
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Yes, line no. 264-67
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	Yes, line no. 268-9
Sponsor	5b	Provide name for the review funder and/or sponsor	N/A
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	N/A
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Yes, line no. 111-25
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Yes, line no. 125-31
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	Yes, line no. 138-65.
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Yes, line no. 167-175
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Yes, table 1
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Yes, line 181-90

Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	Yes, 181-90, and table 2
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Yes, line no. 192-201
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	Yes, table 3
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Yes, line no. 151-65
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Yes, line no. 203-26
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	N/A
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	N/A
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	N/A
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	Yes, line no. 227-43
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	N/A
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	Yes, Line no. 225.

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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